Chapter summary

Familiarising yourself with **items in green** is important for effective revision.
Paying attention to items in the 📝 boxes is helpful for understanding the concept(s).

26.1 Simple Monopoly Pricing

Demand and revenue curves of a monopolist under simple monopoly pricing

- Meaning of simple monopoly pricing
  - **Simple monopoly pricing** refers to a monopolist charging all consumers the same price (a uniform price) for all units of a good.
- The demand curve
  - It is the same as the market demand curve.
- The average revenue curve
  - Total revenue (TR) is equal to price (P) times output (Q).

$$TR = P \times Q$$

  - The average revenue (AR) is equal to the price.

$$\text{Average revenue (AR)} = \frac{\text{Total revenue (TR)}}{\text{Output (Q)}} = \frac{P \times Q}{Q} = P$$

  - The average revenue curve is the same as the downward-sloping market demand curve.
- The marginal revenue curve
  - The marginal revenue (MR) is the change in total revenue resulting from selling an additional unit of output.

$$\text{MR of the } n\text{th unit of output} = \text{TR of selling } n \text{ units of output} - \text{TR of selling } (n-1) \text{ units of output}$$

$$MR = \frac{\text{Change in TR}}{\text{Change in quantity of goods}}$$

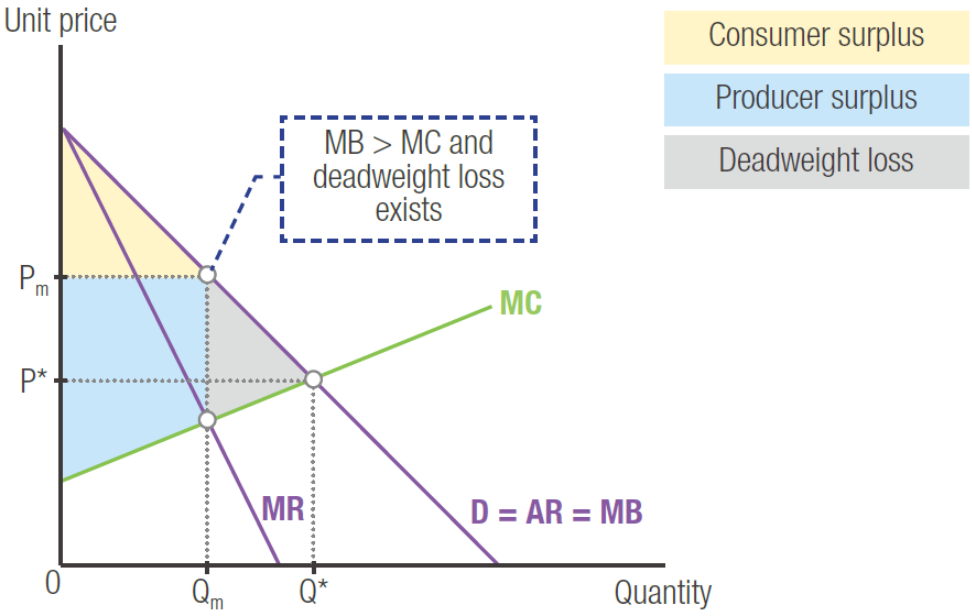
  - Under simple monopoly pricing, when a monopolist wants to sell one more unit of the good, it needs to charge a lower price for all units, not only the additional unit.
  - Under monopoly pricing, **the marginal revenue is smaller than the price and the MR curve lies below the demand curve**.

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Output, price and efficiency implications under simple monopoly pricing

- How does a monopolist determine its output level?
  - A monopolist **produces at an output level where MR = MC** to maximise its profit (Q_m, as show in the figure below).
- How does a monopolist determine its price?
  - A monopolist **sets its price that corresponds to the output level on the demand curve** (P_m, as show in the figure below)
- Efficiency implications under simple monopoly pricing
  - **Efficiency** is attained when total social surplus is maximised (marginal benefit equals marginal cost, i.e. MB = MC)
  - Under simple monopoly pricing, **MB is higher than MC**.
  - There are **potential gains which are not captured by consumers** (the consumer surplus is represented by the yellow area as show in the figure below) **or the monopolist** (the producer surplus is represented by the blue area as shown in the figure below).
  - **There is deadweight loss** (represented by the grey area as show in the figure below).

When there are changes in demand and costs, profit-maximising output and price under simple monopoly pricing may also change.



Comparison between perfect competition and monopoly under simple monopoly pricing

- A monopolist is a price searcher which can change the price of its goods by adjusting its output.
- The market price in a perfectly competitive market is determined by market demand and market supply.

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	Monopoly under simple monopoly pricing	Perfect competition
Output	Monopoly output < Output in a perfectly competitive market	
Price	Monopoly price > Price in a perfectly competitive market	
Marginal revenue	MR < Price	MR = Price
Efficiency	• Consumer surplus is smaller than that in a perfectly competitive market. • Producer surplus may be smaller than, larger than or equal to that in a perfectly competitive market. • Deadweight loss exists. • It is inefficient.	• Consumer surplus is larger than that in a monopoly under simple monopoly pricing. • Producer surplus may be larger than, smaller than or equal to that in a monopoly under simple monopoly pricing. • Deadweight loss does not exist. • It is efficient.

26.2 Price Discrimination

Meaning of price discrimination

- **Price discrimination** is a pricing behaviour whereby **a seller charges different prices for the same good or service to different customers, in which the differences cannot be explained by differences in costs**.

Not all differences in prices are due to practising price discrimination. Differences in prices may simply reflect differences in costs sometimes.

Types of price discrimination

First degree price discrimination	Second degree price discrimination	Third degree price discrimination
• It is also known as perfect price discrimination. • The price of each unit of a good or service is equal to the maximum amount that consumers are will and able to pay. • A seller **charges different buyers different prices depending on their demand for the good or service**.	• It is also known as block pricing or multipart pricing. • A seller **charges different prices for different quantities of a good or service sold**. • The seller does not need to use a lot of effort to assess buyers' information.	• This type of price discrimination is sometimes referred to as market segmentation. • **Different prices are charged in different market segments with different elasticities of demand for the same good produced at the same cost**. • Markets with more (less) elastic demand are usually charged with lower (higher) prices.

Some economists argue that block pricing is not a kind of price discrimination, as long as all customers face the same price schedule and there is no market segmentation.

Price and output determination and efficiency implications of different types of price discriminations are not required in the curriculum.

Conditions for practising price discrimination

- Sellers are price searchers
- Separable markets
- Different price elasticities of demand
- Different information costs

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